Data Appendix

Dataset: uva_reviews_with_sentiment.csv

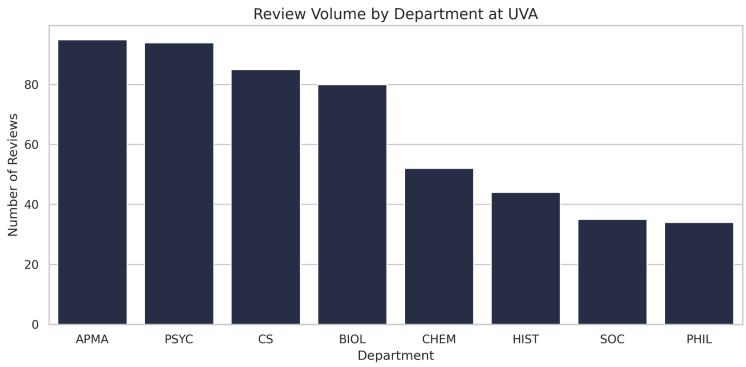
Unit of Observation: The unit of observation in this dataset is an individual review of a specific undergraduate UVA course scraped from theCourseForum. Each review contains the review text, course, date, rating, and the sentiment analysis scores.

Variable: Department

- Definition: 2-4 letter academic code representing the department offering the course.
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv), which was originally scraped from theCourseForum website.
- Missing observations: 519(0)
- Frequency table:
  -

Department	Frequency	Percentage
APMA	95	18.3
PSYC	94	18.1
CS	85	16.4
BIOL	80	15.4
CHEM	52	10.0
HIST	44	8.5
SOC	35	6.7
PHIL	34	6.6

- Visualization:

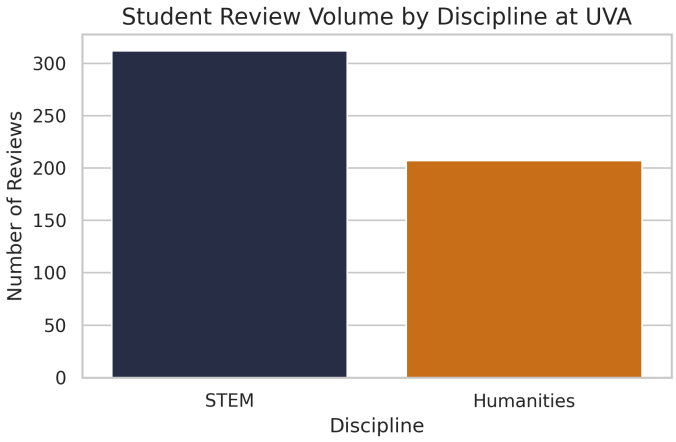


Variable: Group

- Definition: The broad disciplinary category assigned to the department for the purpose of this analysis.
- Source and construction: Constructed by mapping the ‘department’ variable to either ‘STEM’ or ‘Humanities’. “CS”, ”BIOL”,”CHEM”, “APMA” were assigned as “STEM”; “SOC”, ”PHIL”, “HIST”, and “PSYC” were assigned to ‘Humanities’
- Missing observations: 519(0)
- Frequency table:

Group	Frequency	Percentage
STEM	312	60.1
Humanities	207	39.9

- Visualization:

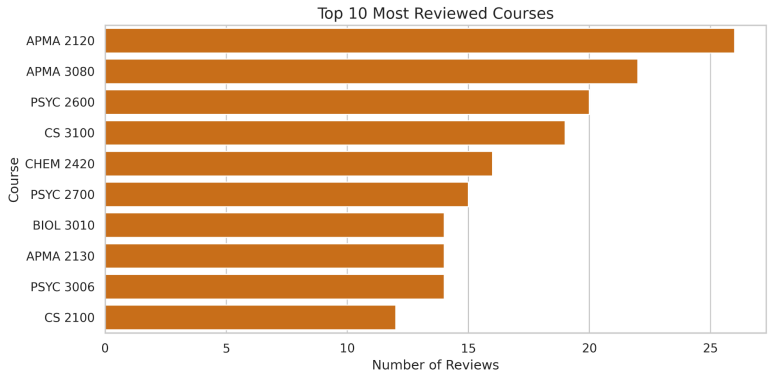


Variable: Course

- Definition: The specific course mnemonic and number for the course being reviewed
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv) from the column ‘course’.

- Missing observations: 519(0)
- Frequency table:
  - Top 5 Most Reviewed Courses:
  -

Course	Frequency	Percentage
APMA 2120	26	5.0
APMA 3080	22	4.2
PSYC 2600	20	3.9
CS 3100	19	3.7
CHEM 2420	16	3.1

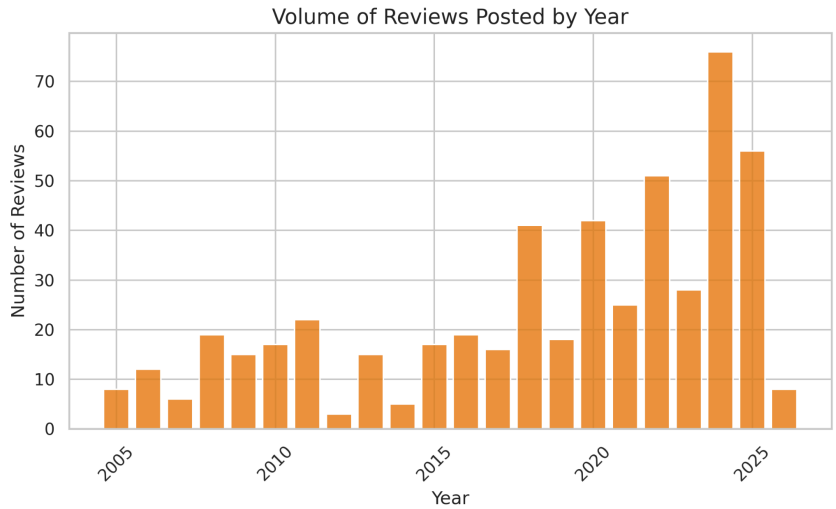
- Visualization:
  - 

Variable: Date

- Definition: Date that the review was originally posted or last updated on theCourseForum website.
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv) under the column ‘date’.
- Missing observations: 519(0)
- Frequency table:
  -

Year	Frequency	Percentage
2026	8	1.5
2025	56	10.8
2024	76	14.6
2023	28	5.4

2022	51	9.8
2021	25	4.8
2020	42	8.1
Pre-2020	233	44.9

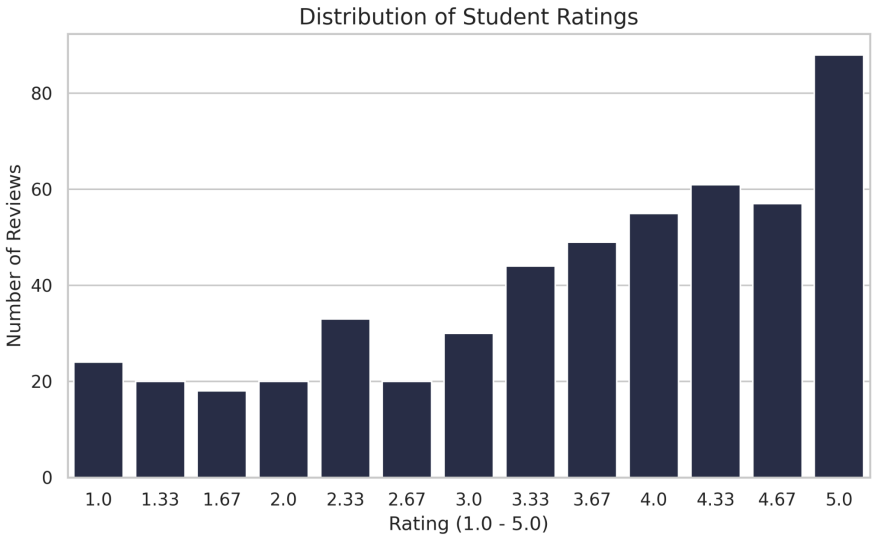
- Visualization:
  - 

Variable: Rating

- Definition: Reported numeric score given by students for the course as part of their review (from 1.0-5.0).
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv) under the column ‘rating’.
- Missing observations: 519(0)
- Summary statistics:
  -

Count	519
Mean	3.58
Standard Deviation	1.20
Minimum	1.00
25%	2.67
50%	4.00
75%	4.67
Maximum	5.00

- Visualization:

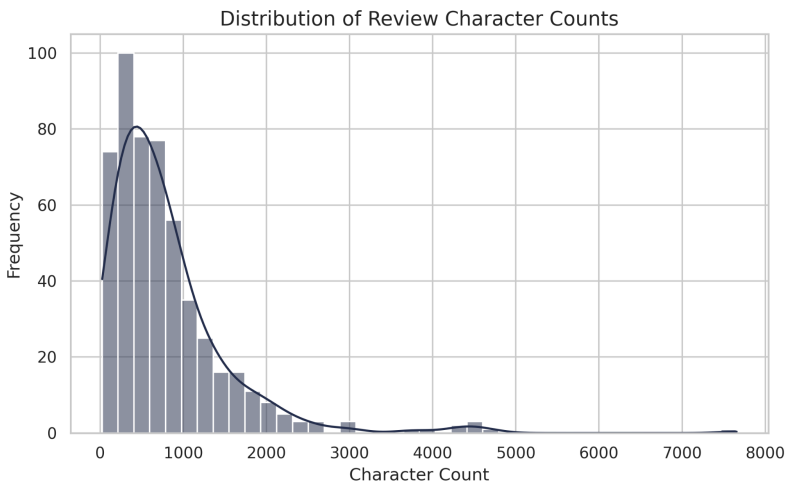


Variable: Text

- Definition: Character count of the student's text review for the course.
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv) under the column 'text'. Reviews of less than 20 characters were excluded.
- Missing observations: 519(0)
- Summary statistics:

Count	519
Mean	798
Standard Deviation	765
Minimum	24
25%	325
50%	616
75%	992
Maximum	7655

- Visualization

- 

Variable: URL

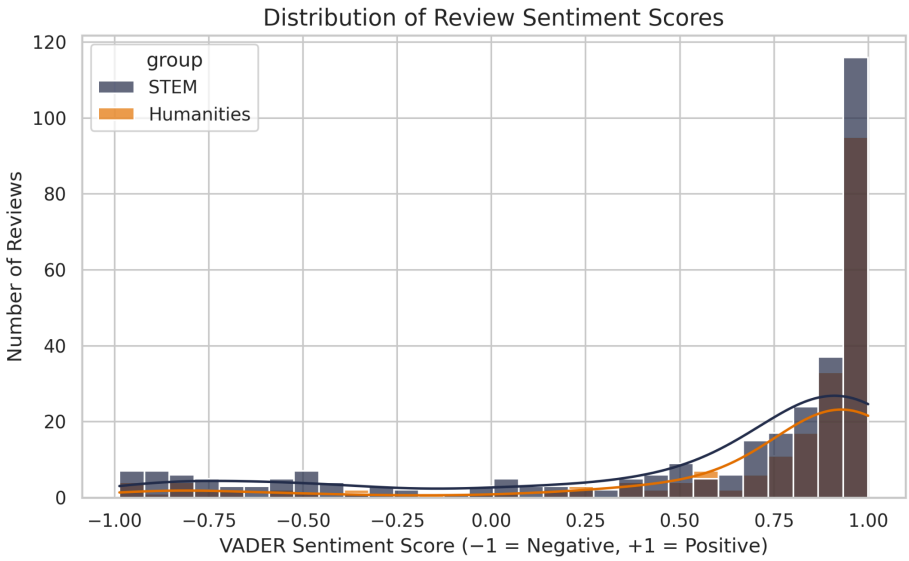
- Definition: The web address of the course page from where the review was found.
- Source and construction: Extracted directly from the input data file (uva_reviews_final.csv) under the column ‘url’.
- Missing observations: 519(0)
- Summary statistics: Unique identifier, no statistics available

Variable: Sentiment Score

- Definition: The compound sentiment score of the user’s review based on their language used, ranging from -1.0 to 1.0.
- Source and construction: Constructed by applying VADER algorithm to ‘text’ variable from the input data file (uva_reviews_final.csv).
- Missing observations: 519(0)
- Summary statistics:
  -

Count	519
Mean	0.6158
Standard Deviation	0.5678
Minimum	-0.9872
25%	0.5414
50%	0.8945
75%	0.9719
Maximum	0.9996

- Visualization:

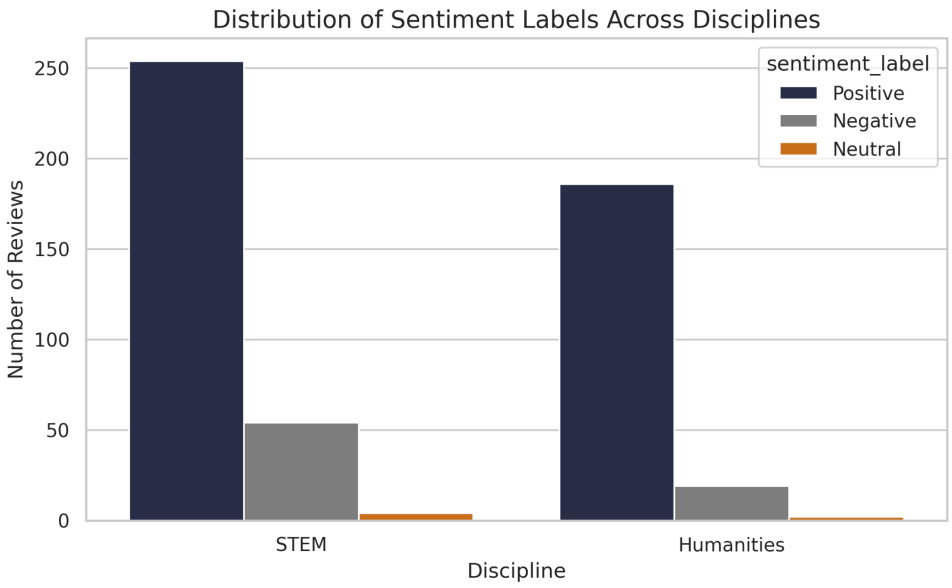


Variable: Sentiment Label

- Definition: The classification of the calculated sentiment score into ‘Positive’, ‘Neutral’, ‘Negative’ labels.
- Source and construction: Constructed by assigning values based on the sentiment score: “Positive” - score >= 0.05, “Negative” - score <= -0.05, else “Neutral”.
- Missing observations: 519(0)
- Frequency Table:

Label	Frequency	Percentage
Positive	440	84.8
Negative	73	14.1
Neutral	6	1.2

- Visualization:


Distribution of Sentiment Labels Across Disciplines
sentiment_label
Positive
Negative
Neutral
Number of Reviews
250
200
150
100
50
0
STEM
Humanities
Discipline